



OFFICIAL OPERATIONAL HANDBOOK & USER MANUAL

# NDLI Club Management & Regional Employee Activity Tracking System

A Unified, Enterprise-Grade Operational Guidebook for Central Administrative Governance, 7 Regional Nodal Zone Workspaces, Automated 7-Day Backups, and Real-Time Multi-State Activity Tracking.

- ✓ Dual-Portal Architecture
- ✓ 7 Regional Nodal Zones
- ✓ 36 States/UTs Automated Mapping
- ✓ 50/30/20/10 Star Rating Matrix
- ✓ AI Strategic Decision Engine
- ✓ Automated 7-Day Rolling Backups
- ✓ Dynamic Renewal Certificates

DOCUMENT RELEASE  
**Production Edition v2.4 • September 2026**  
SYSTEM DEPLOYMENT TARGET  
**Cloud Web Service & Google Drive / GAS Hybrid Architecture**

**LEAD SYSTEM ARCHITECT & FULL-STACK DEVELOPER**  
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Indian Institute of Technology Kharagpur

# 1. Preface

The **National Digital Library of India (NDLI)** Club initiative, coordinated by the **Indian Institute of Technology Kharagpur (IIT Kharagpur)** under the Ministry of Education, Government of India, represents a nationwide endeavor to empower educational institutions, cultivate lifelong reading habits, and bridge the digital knowledge divide across all 36 States and Union Territories.

As the network of authorized NDLI Clubs expanded exponentially into thousands of secondary schools, universities, and research institutes, managing field engagements and institutional verifications necessitated an enterprise-grade, resilient software platform. The **NDLI Club Management and Employee Activity Tracking System** was architected and developed by **Dr. Anirban Mukherjee** to serve as this centralized operational bridge.

## 1.1 Core Architectural Principles

- **Zero External Dependency Kernel:** Engineered entirely on the native Python standard library (HTTP server, JSON, and CSV engines). Runs instantly on any operating system without complex runtime dependency conflicts.
- **Dual-Portal Governance:** Logical and physical separation between the Central Administration Office Master Portal at IIT Kharagpur and the 7 Regional Zonal Workspaces distributed across India.
- **Node-to-Master Synchronization:** Regional officers log field activities and club approvals into their dedicated node CSV files, which are immediately reconciled into the central master databases with strict atomic locking.
- **Autonomous Data Resilience:** Automated rolling backups execute every 7 days, maintaining a strict 2-version retention policy to guarantee deterministic storage bounds and data safety.

## 1.2 Scope & Intended Audience

This operational handbook provides authoritative guidance for Central Administrative Officers at IIT Kharagpur, the 7 Regional Zonal Officers managing daily zonal operations, and DevOps engineers overseeing cloud deployments.

### Institutional Attribution & Brand Identity

The application adheres strictly to the official IIT Kharagpur identity, utilizing deep forest green tones ( #103125 , #1A503E , #0D7751 ) and verified developer attribution to Lead Architect **Dr. Anirban Mukherjee** across all portals, verification certificates, and administrative endpoints.

## 2. Feature Highlights & Executive Overview

The platform integrates an extensive suite of administrative governance and regional execution capabilities:

**Central Administration Dashboard**

Real-time KPI telemetry, Donut and Trajectory charts, Renewal Attention centers, and multi-dimensional scope filtering by Zone and Academic Year.

**Employee Performance Star Matrix**

Statutory 50/30/20/10 weighted formula evaluating regional officers with dynamic 1-to-5 star rating badges, filtered by Year, Month, and State.

**National Penetration Roster**

Exhaustive roster covering all 36 States & UTs with real-time club density, recent activity timestamps, and responsive no-horizontal-scroll layout.

**AI Strategic Decision Engine**

Heuristic intelligence formulating policy recommendations, identifying penetration deficits, and generating 30/90/365-day milestone roadmaps.

**Regional Daily Activity Logging (SEC A)**

Real-time logging across 4 categories: Phone/Remote, OS Ticket Closing, Online Webinars, and Offline Field Training with automatic ISO timestamps.

**Club Approval & Auto-Zone Engine (SEC B/C)**

Standardized onboarding capturing institutional stakeholders and automatically mapping selected states across 36 territories to their 6 statutory zones.

**Universal Search & Metadata Editor**

Instant live-indexed search across Club ID, Registration No, State, or Institute name with in-place record correction and atomic master synchronization.

**Dynamic Verification Certificates**

Generates official IIT Kharagpur Renewal Certificates embedding official seals, dynamic signatories, Sutradhar typography, and instant vector PDF export.

**Master DB Reconciliation Engine**

1-click audit scanning all 7 regional nodes, recovering missing entries, recalibrating quotas, and ensuring 100% data integrity without data loss.

**Google Drive Cloud Mirroring**

Automated cloud mirroring of 7-day and manual ZIP backups directly into Google Drive with strict 2-version rolling retention.

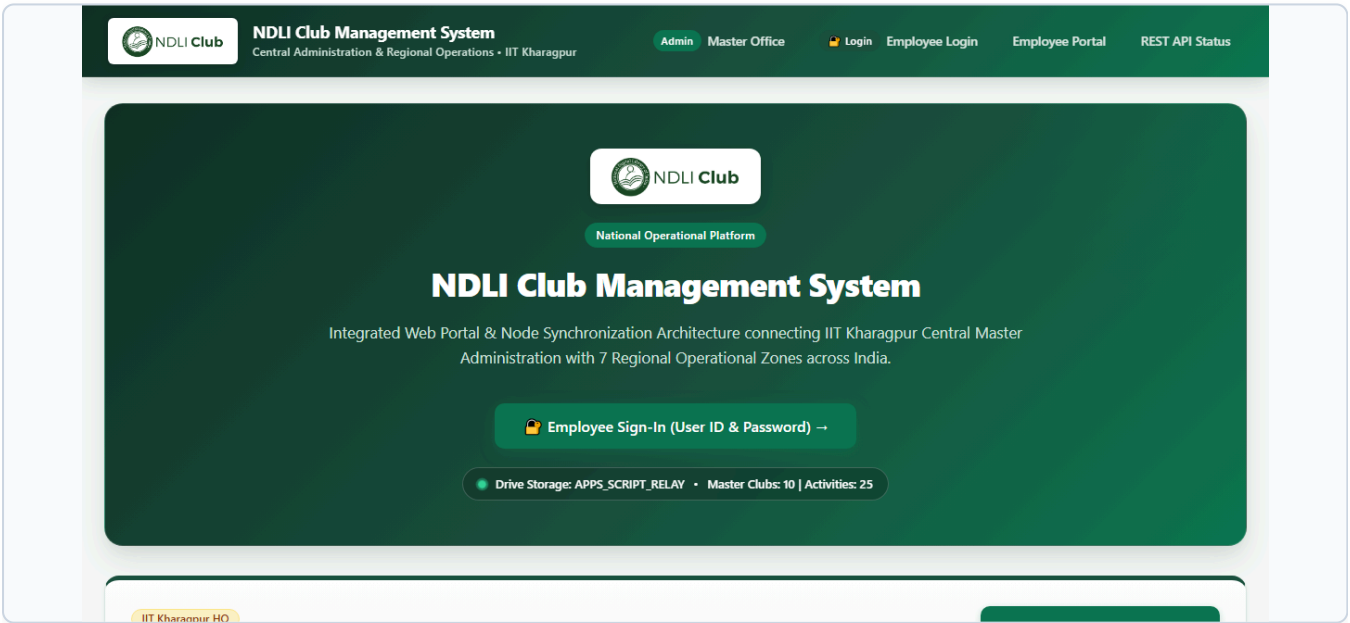


Figure 2.1: Unified Landing Portal — Dual Gateway for Central Administration and 7 Regional Zonal Workspaces

### 3. Table of Contents & Navigation Index

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## 4. Detailed Specifications & Schemas

The platform is powered by a high-performance, atomic CSV database engine. Every operation is schema-validated to guarantee consistency across master and employee node tiers.

### 4.1 Master Database Schema Specifications

| Database File         | Schema Columns                                                                                                                             | Operational Function                                                                                                         |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| master_clubs.csv      | id, reg_no, name, state, zone, patron_email, president_email, secretary_email, approved_by, approval_date, last_renewal_date, renewal_date | Stores all active NDLI Clubs nationwide. Enforces strict +1 year renewal date invariants from establishment or last renewal. |
| master_activities.csv | id, emp_id, timestamp, support_type, notes, state, zone                                                                                    | Aggregated log of all field interactions (calls, ticket closures, webinars, offline campus visits) across India.             |
| master_quotas.csv     | emp_id, emp_name, zone, clubs_approved, total_activities, last_activity, updated_at                                                        | Scoreboard tracking individual employee performance quotas and daily logging volume.                                         |
| master_users.csv      | id, email, password_hash, salt, full_name, role, zone, assigned_states, is_active, created_at                                              | Credentials database. Passwords secured using PBKDF2-HMAC-SHA256 (100,000 rounds) with per-user 16-byte random salts.        |
| master_issues.csv     | id, club_id, club_name, emp_id, reported_at, issue_notes, status, next_reminder_date, resolved_at, resolution_notes                        | Engine for 72-hour employee reminders, 7-day recurrent cycles, and 30-day Admin escalations.                                 |

### 4.2 Regional Zonal Jurisdiction Roster

| User ID | Regional Officer   | Zone       | Territorial Jurisdiction (36 States & UTs)                                                                |
|---------|--------------------|------------|-----------------------------------------------------------------------------------------------------------|
| EMP01   | Rohan Sharma       | NORTH      | Jammu & Kashmir, Ladakh, Uttarakhand, Himachal Pradesh, Chandigarh, Punjab, Haryana, Delhi, Uttar Pradesh |
| EMP02   | Pooja Verma        | CENTRAL    | Madhya Pradesh, Chhattisgarh                                                                              |
| EMP03   | Amit Patel         | WEST       | Rajasthan, Gujarat, Maharashtra, Goa, Daman and Diu, Dadar & Nagar Haveli                                 |
| EMP04   | Debabrata Ghosh    | EAST       | Bihar, Jharkhand, West Bengal, Odisha                                                                     |
| EMP05   | Mayanglambam Singh | NORTH EAST | Sikkim, Assam, Arunachal Pradesh, Meghalaya, Manipur, Tripura, Nagaland, Mizoram                          |
| EMP06   | K. Venkatesh       | SOUTH      | Andhra Pradesh, Telangana, Karnataka (South Sector 1)                                                     |
| EMP07   | Ananya Nair        | SOUTH      | Tamil Nadu, Puducherry, Kerala, Andaman & Nicobar Island, Lakshadweep (South Sector 2)                    |

## 5. Regional Employee Portal Operations

The **NDLI Club Employee Portal** is the operational cockpit for the 7 Regional Zonal Officers. It features real-time clock synchronization, pending reminder feeds, and structured data entry workflows.

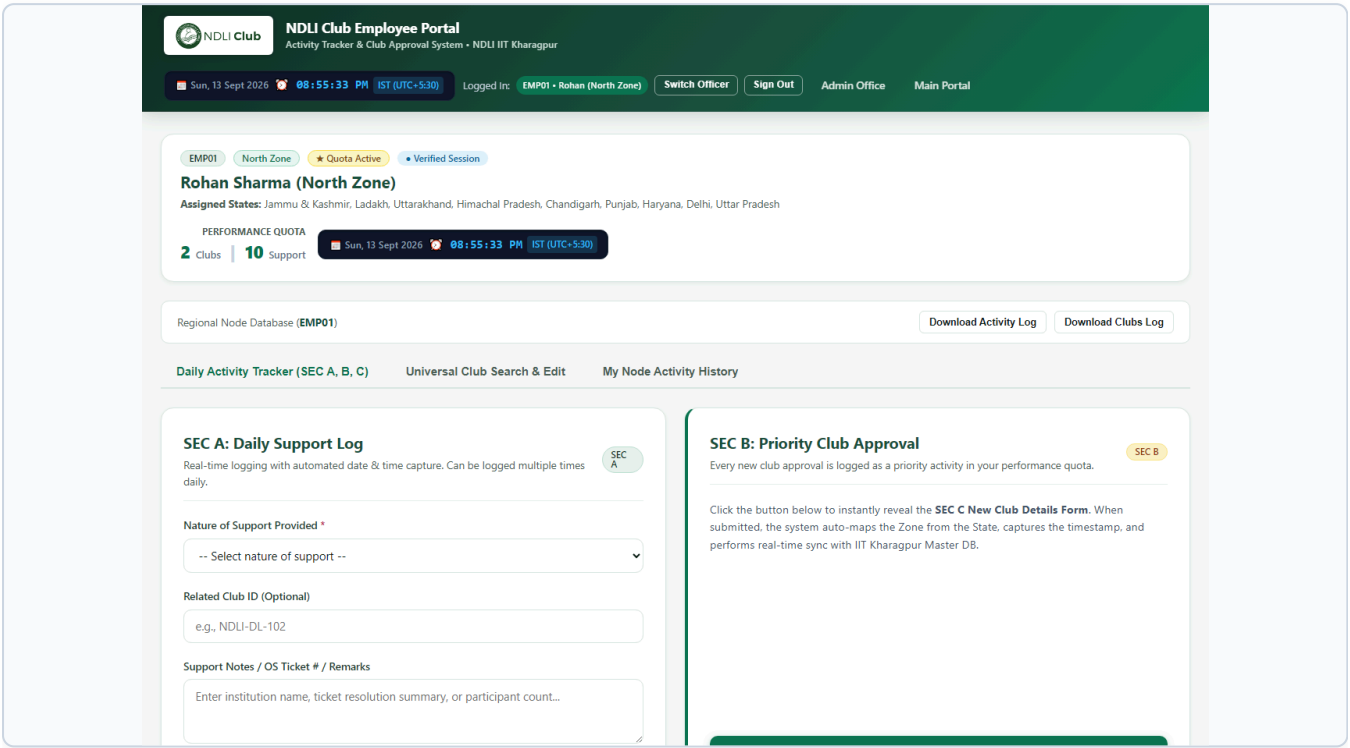


Figure 5.1: Regional Employee Workspace (EMP01 - Rohan Sharma, North Zone)

### 5.1 Logging In & Selecting Zonal Officer Profile

1. **Navigate to Portal:** Open `/employee` or select "NDLI Club Employee Portal" from the landing gateway.
2. **Interactive Officer Directory:** Click any of the 7 regional officer chips (e.g., *EMP01 - Rohan Sharma (North)*). The system automatically inputs their official email and pre-configures their territorial scope.
3. **Password Authentication:** Enter the assigned regional password (e.g., `EmpNorth#2026`) and click "Authenticate & Open Portal".
4. **Workspace Activation:** An encrypted 24-byte session token is generated, opening the officer's dedicated workspace with their node database pre-loaded.

## 5.2 Step-by-Step: Logging Daily Support Activities (SEC A)

Zonal officers engage with educational institutions daily to resolve obstacles and conduct orientation sessions. Every interaction must be logged immediately in **SEC A**:

1. **Select Nature of Support:** Choose the appropriate activity tier:
  - **Phone call and remote assistance:** Voice calls, WhatsApp consultations, and remote desktop setup.
  - **Closing of OS Ticket:** Formal resolution of institutional support tickets logged via NDLI helpdesk.
  - **Online training:** Conducting virtual orientation webinars, faculty workshops, and student onboarding.
  - **Offline training:** On-premise physical campus visits, seminars, and library workshops.
2. **Activity Notes:** Enter brief notes identifying the institution, key stakeholders, and topics addressed.
3. **Submit Entry:** Click "**Submit Activity Log**". The entry is recorded in `activity_log.csv` with automatic timestamping and synced incrementally to the Central Master Database.

## 5.3 Step-by-Step: Approving & Onboarding New NDLI Clubs (SEC B & C)

When an institution completes its registration requirements, the officer authorizes the club through **SEC B & C**:

1. **Trigger Approval Form:** Click the button "**⚡ SEC B: Approve New NDLI Club**" to reveal SEC C.
2. **Input Institutional Codes:** Enter the unique Club ID (e.g., `CLUB-DEL-2026-01`), Registration Number, and Institution Name.
3. **State Selection (Auto-Zone Engine):** Select the institution's state from the dropdown. The system automatically determines and locks the statutory Zone (North, Central, West, East, North East, South), preventing territorial misclassification.
4. **Stakeholder Contact Matrix:** Provide verified email addresses for Patron (Head of Institution), President (Faculty Lead), and Secretary (Student Representative).
5. **Authorize Club:** Click "**Confirm & Authorize New NDLI Club**". The club is committed to the node database, increments the officer's performance quota, and syncs to `master_clubs.csv`.

### ⚠ **Mandatory Next Renewal Date Invariant (Rule 10)**

Upon club approval, the system enforces **Rule 10**: the upcoming renewal date is automatically set to *exactly 1 calendar year (+1 year)* from the Date of Approval (DOA). Under no circumstances can a renewal date precede DOA.

## 5.4 Universal Club Search, Record Editing & Annual Renewal

The lower section of the Employee Portal provides comprehensive directory lookup and annual renewal workflows:

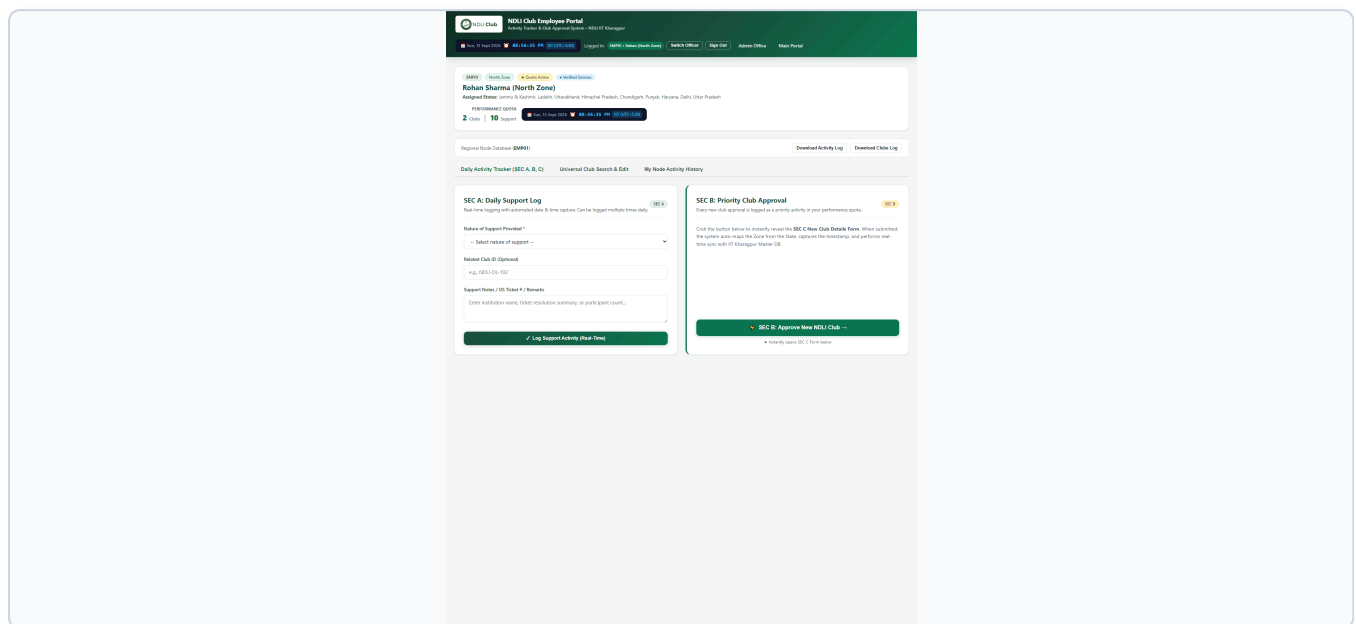


Figure 5.2: Employee Portal Lower Workspace — Universal Search, Details Inspector, and 72-Hour Reminder Stream

1. **Universal Keyword Query:** Enter any search string into the search box. The table dynamically filters by Club ID, Registration No, State, or Institute name.
2. **Inspect & Edit Metadata:** Click "**Inspect / Edit**" on any club card. An interactive modal displays all current parameters, allowing officers to update stakeholder emails and institutional names with instant master sync.
3. **Execute Annual Renewal:** Click "**Process Registration Renewal**", verify credentials, and confirm. The system updates the Last Renewal Date (LRD) to today, calculates the next renewal date as +1 year, and generates a downloadable verification certificate.

## 5.5 Unresolved Club Issues & 72-Hour Reminders

When an institution reports an unresolved issue, the officer clicks "**Report Issue / Set Reminder**":

- **72-Hour Grace Window:** The ticket enters a pending state. Exactly 72 hours later, a prominent reminder banner appears in the employee's dashboard header.
- **Resolution Workflow:** Clicking the reminder allows the officer to mark it "**Resolved**" or "**Not Resolved**". If unresolved, the system automatically reschedules a reminder every 7 days until resolved.
- **Exporting Node CSVs:** Officers can click "**Download Activity Log**" and "**Download Clubs Log**" to export exact copies of their node databases for local record-keeping.

## 6. Central Administration Office Portal Operations

The **Central Administration Office Portal** at IIT Kharagpur is the executive master control console. It synthesizes field reports from all 7 regional zones into actionable statistical intelligence.

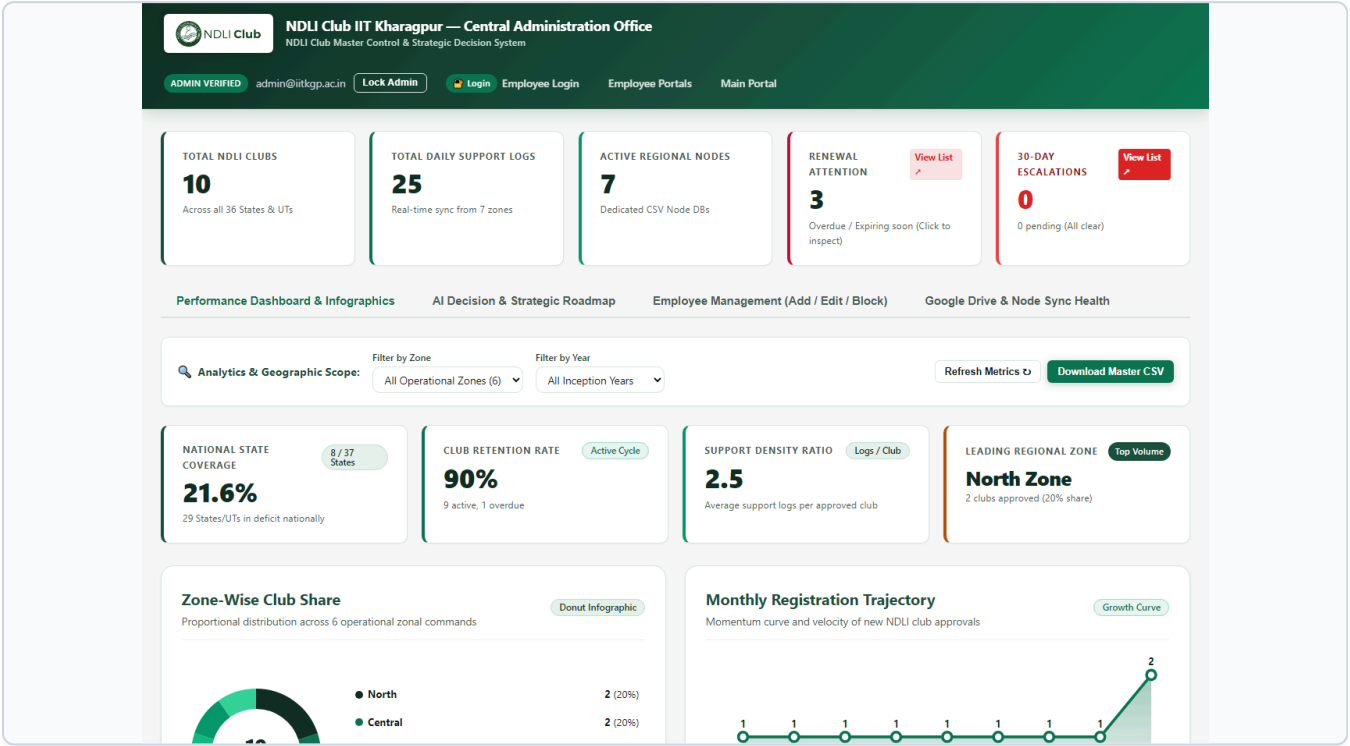


Figure 6.1: Central Administration Executive Dashboard — Real-Time KPIs, Donut Charts & Scope Filters

### 6.1 Admin Security & Executive KPI Monitors

Administrative access is protected by role-based authentication ( admin@iitkgp.ac.in ). The dashboard displays five top-level operational meters:

- **Total NDLI Clubs:** Cumulative authorized clubs across all 36 States & UTs.
- **Total Daily Support Logs:** Nationwide count of field interactions submitted across all regional nodes.
- **Active Regional Nodes:** Number of operational regional databases communicating in real-time.
- **Renewal Attention Center:** Highlights clubs with expired or imminent registration deadlines (within 30 days). Clicking the card opens an interactive drill-down inspection list.
- **30-Day Escalations Monitor:** Displays persistent club issues unresolved for 30 or more days across any zone, requiring central administrative intervention.

## 6.2 Statistical Infographics & Geographic Scope Filtering

The dashboard renders dynamic visual charts representing national adoption patterns:

- **Zone-Wise Distribution Donut Chart:** Displays institutional concentration across the 6 geopolitical zones (North, Central, West, East, North East, South).
- **Monthly Trajectory Area Chart:** Illustrates onboarding velocity month-over-month, showing seasonal registration trends.
- **Support Classification Breakdown:** Tracks the nationwide proportion of remote phone support, OS ticket resolutions, online webinars, and physical campus trainings.
- **Scope Filter Bar:** Administrators can filter all dashboard metrics simultaneously by **Zone** (e.g., "North Zone Only") and **Academic Year** (e.g., 2026, 2025).

## 6.3 Handling 30-Day Administrative Escalations

Issues unresolved at the regional level for more than 30 days automatically trigger central escalation:

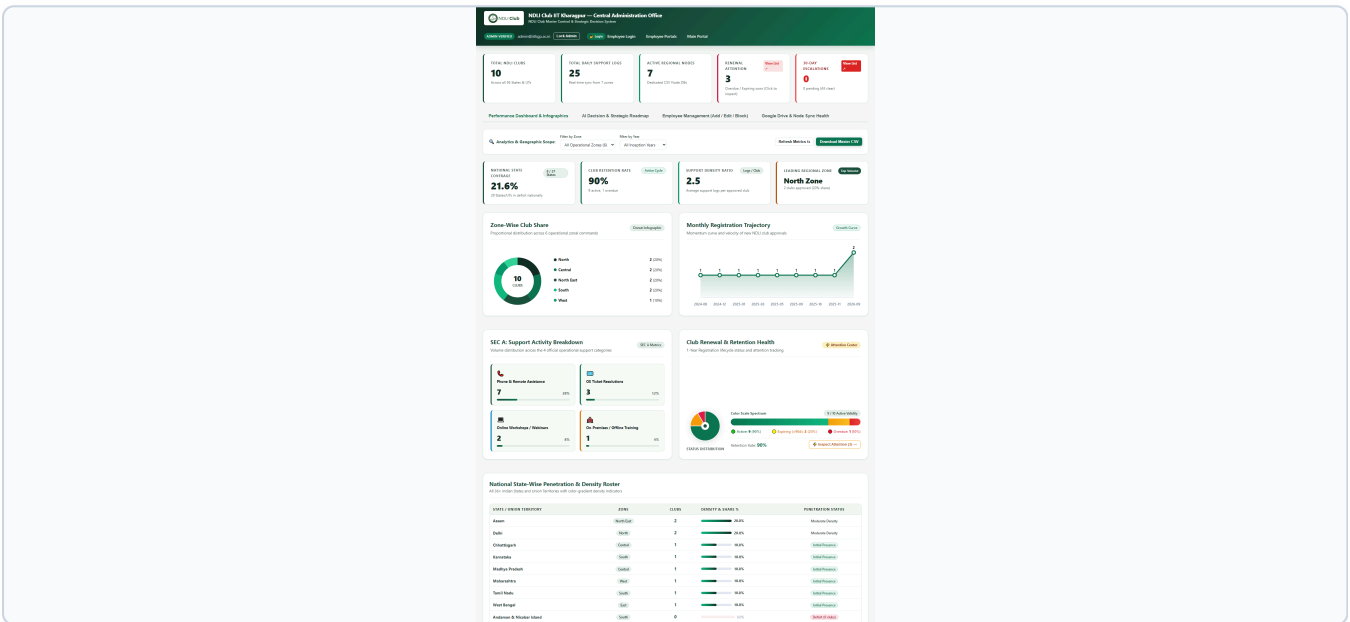


Figure 6.2: Admin Portal Lower Section — Penetration Roster, Employee Performance Matrix & Escalation Monitors

1. Click the red **"30-Day Escalations"** card in the KPI header.
2. Review the institution's grievance history, originating zonal officer, and days elapsed.
3. Click **"Review & Resolve"** to record Central Office directives and mark the ticket **"Resolved"** or **"Not Resolved"** (re-queuing for 7-day central tracking).

## 6.4 Employee Performance Evaluation & Star Rating Matrix

The **Employee Performance Evaluation & Star Rating Matrix** provides an objective merit evaluation of all 7 regional zonal officers based on an approved multi-criteria weighting formula:

$$\text{Composite Score} = (\text{Clubs Approved} \times 0.50) + ((\text{OS Tickets Closed} + \text{Phone Calls}) \times 0.30) + (\text{Offline Training} \times 0.20) + (\text{Online Training} \times 0.10)$$

| Evaluation Metric                | Weightage  | Operational Significance                                                                  |
|----------------------------------|------------|-------------------------------------------------------------------------------------------|
| New Club Approvals               | 50% WEIGHT | Primary organizational KPI reflecting institutional expansion and onboarding capacity.    |
| OS Tickets Closed & Remote Calls | 30% WEIGHT | Measures responsiveness to club operational queries and problem-solving velocity.         |
| Offline Field Training           | 20% WEIGHT | Physical campus workshops, library orientations, and direct faculty development sessions. |
| Online Training                  | 10% WEIGHT | Virtual webinars, digital orientation sessions, and remote demonstrations.                |

### Star Rating Thresholds & Categorization

| Star Rating | Normalized Benchmark    | Performance Appraisal Designation                                                                                  |
|-------------|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| ★★★★★       | 80% – 100% of Benchmark | <b>Outstanding:</b> Consistently exceeds institutional quotas, fast ticket resolution, proactive field engagement. |
| ★★★★        | 60% – 79% of Benchmark  | <b>Commendable:</b> Strong performance across onboarding, ticket closing, and training.                            |
| ★★★         | 40% – 59% of Benchmark  | <b>Satisfactory:</b> Fulfills standard institutional quotas and baseline activity targets.                         |
| ★★          | 20% – 39% of Benchmark  | <b>Needs Improvement:</b> Below expected club onboarding velocity or lagging ticket resolution.                    |
| ★           | 0% – 19% of Benchmark   | <b>Critical Deficit:</b> Immediate administrative consultation and field support required.                         |

### Dynamic Analytical Filtering

Administrators can filter performance records by **Year** (e.g., 2026), **Month** (January–December), and **State** to evaluate seasonal performance or regional specializations.

## 6.5 National State-Wise Penetration & Density Roster

The **National State-Wise Penetration & Density Roster** renders all 36 States and Union Territories with zero horizontal scrolling, enabling instant nationwide assessment:

- **Territory & Jurisdiction:** Lists each State/UT, assigned Regional Zonal Officer, and geopolitical Zone.
- **Club Density:** Real-time count of active NDLI Clubs operating within that state.
- **Activity Timestamp:** Displays the exact date of the latest onboarding or support engagement.
- **Health Status Badges:** Color-coded indicators denoting High Density (Green), Medium Density (Gold), or Priority Expansion (Red).

## 6.6 AI Strategic Decision Module & Policy Roadmaps

The **AI Strategic Decision Engine** executes continuous heuristic evaluation across the aggregated master databases to generate strategic recommendations for central leadership:

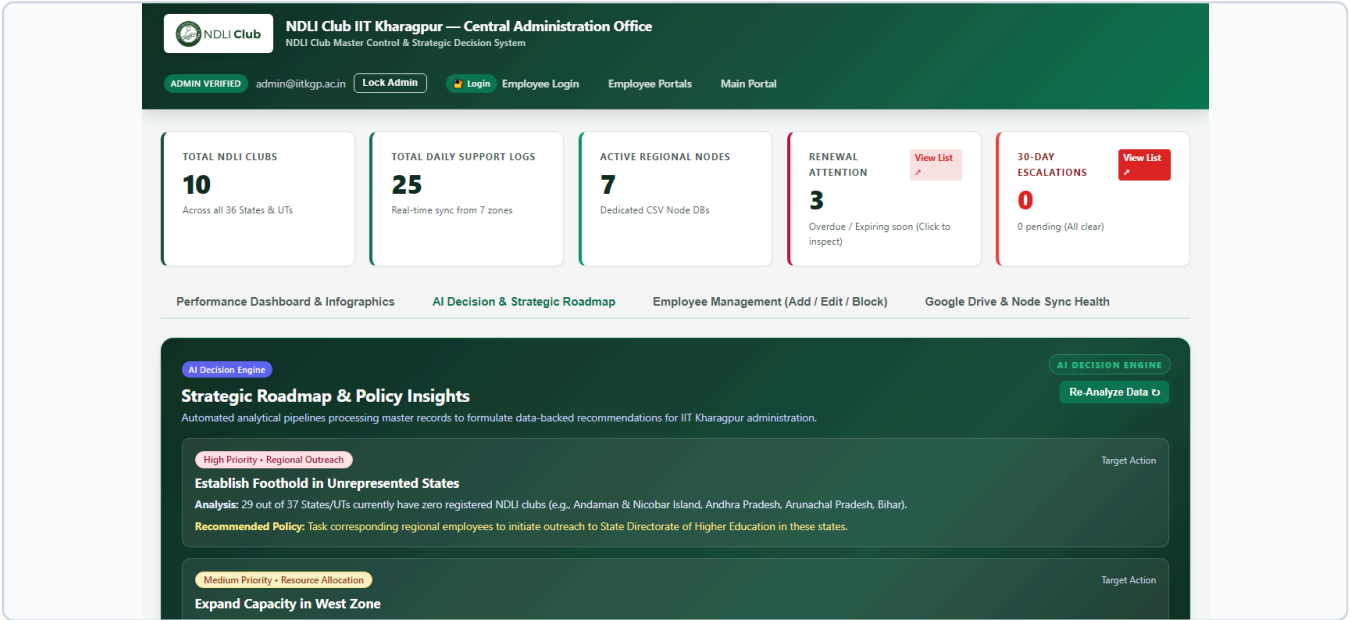


Figure 6.3: AI Strategic Decision Engine — Automated Growth Corridors, Deficit Analysis & Milestone Roadmaps

- **State Penetration Deficit Formulation:** Automatically identifies territories with club densities below national population-adjusted benchmarks, recommending targeted state-level campaigns.
- **High-Growth Corridor Detection:** Identifies zones exhibiting accelerating adoption momentum to allocate supplemental training materials and regional resources.
- **Milestone Policy Roadmaps:** Synthesizes operational directives into **30-Day Immediate**, **90-Day Mid-Term**, and **1-Year Strategic Milestones**.

## 6.7 Node-to-Master Database Reconciliation Engine

The **Master DB Reconciliation** module (operated via the "**Run Full Node-to-Master Reconciliation**" button in the Admin Office Portal) serves as the primary system-wide data integrity and state recovery mechanism for IIT Kharagpur:

- **Distributed Node Audit:** Recursively scans all regional officer node databases across India ( data/employees/emp01/ through emp07/ , plus any newly provisioned officers), inspecting their local clubs.csv and activity\_log.csv files.
- **Missing Record Discovery & Merge:** Discovers any club approvals or support activities logged during intermittent network connectivity or restored from historical backups, and seamlessly merges them into the Central Master Database without overwriting newer records or losing data.
- **Schema Normalization & Renewal Recalculation:** Validates all date stamps against strict ISO-8601 formatting, verifies stakeholder contact emails, and recalculates dynamic annual club renewal dates.
- **Quota & Star Rating Synchronization:** Recomputes each officer's approved club counts and support activity tallies directly from raw source records, guaranteeing that master\_quotas.csv and the **Employee Performance Evaluation & Star Rating Matrix** display 100% verified, tamper-proof ratings.
- **Operational Best Practice:** Central administrators should trigger full reconciliation after restoring backups, after manual CSV imports/edits, or whenever an administrative data audit is conducted.

## 6.8 Automated 7-Day Backups, Cloud Mirroring & One-Click Emergency Restore

The platform incorporates an autonomous background backup engine and a disaster recovery console in the Admin Portal:

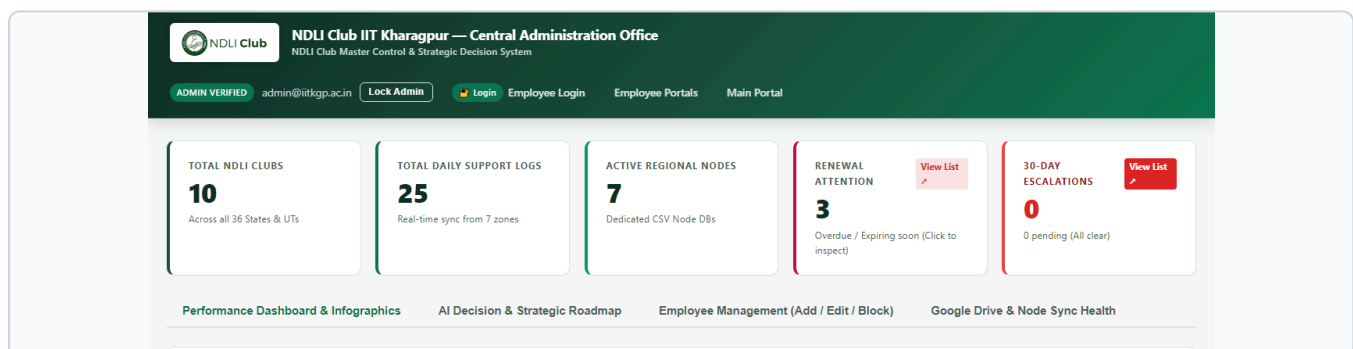


Figure 6.4: Master DB Reconciliation Console, 7-Day Rolling Backup Engine & Google Drive Cloud Mirror

- **Autonomous 7-Day Cadence:** Archives both Central Master DBs and all 7 Regional Node DBs into compressed ZIP snapshots.
- **Strict 2-Version Retention Invariant:** Preserves only the last 2 historical snapshots. All older archives are automatically pruned, preventing storage bloat.
- **Google Drive Webhook Relay:** Every backup automatically mirrors a timestamped ZIP archive directly into the Google Drive backups/ repository via Webhook Relay.
- **One-Click Emergency Database Restoration:** Administrators can roll back the entire system from retained snapshots or Google Drive archives via the "**One-Click Restore Database** ↺" button under *Google Drive & Node Sync Health*.
- **Two-Layer Security Protocol:** To prevent accidental or unauthorized triggers, restoration requires solving a randomized 5-character alphanumeric anti-robot CAPTCHA code and re-verifying the Central Admin password.
- **Automated Post-Restore Reconciliation:** Following archive decompression, the engine immediately runs full node-to-master reconciliation, ensuring 100% data consistency and updated ratings across all nodes.

## 6.9 Dynamic Verification Certificate Customization

Administrators can configure the layout and visual assets of official NDLI Club Verification Certificates, including authorized signatory designations, high-resolution PNG signatures, and the official IIT Kharagpur institutional seal. Every

## 7. Maintenance, Cloud Deployment & FAQs

### 7.1 File & Directory Topology

```
ndli_club_management/
├── app.py                # HTTP Server & REST API Controller
├── auth.py              # PBKDF2 Password & Session Manager
├── config.py            # Global Ports & System Constants
├── state_zone_mapper.py # 36 States/UTs to 6 Zones Mapping Rules
├── ai/
│   └── decision_module.py # AI Strategic Analysis Engine
├── db/
│   ├── csv_engine.py    # Atomic CSV Engine with Mutex Locking
│   ├── sync_engine.py   # Node-to-Master Synchronization Core
│   ├── backup_engine.py # 7-Day Scheduler & Retention Safeguard
│   └── issue_manager.py  # 72-Hour Reminders & 30-Day Escalations
├── data/
│   ├── master/          # Central Master CSV Databases
│   ├── employees/       # 7 Regional Node DBs (emp01-emp07)
│   └── backups/          # Automated Rolling ZIP Archives (Max 2)
├── templates/           # Production Frontends (admin, employee, index)
└── static/              # CSS Stylesheets, Branding Assets & Scripts
```

### 7.2 Render Cloud Deployment Specification

- Environment:** Python 3.10+ (Standard Library, zero external dependencies).
- Build Command:** `python -c "print('Build verified')"` (or leave blank).
- Start Command:** `python app.py`
- Port Binding:** Dynamically reads `$PORT` environment variable (defaults to 8080).
- Health Check Endpoint:** `/api/health` (returns HTTP 200 OK with service status).

### 7.3 Frequently Asked Questions (FAQs)

#### Q: How do regional entries sync with the master database?

When an officer logs an activity or approves a club, the sync engine writes to the local node CSV and immediately updates the master CSV with atomic mutex locking, rendering on the admin dashboard within seconds.

#### Q: What happens if an employee experiences intermittent internet connectivity?

The web client holds pending submissions in memory and retries upon reconnection. The backend engine prevents duplicate record creation using unique Club ID indexes.

#### Q: What is the function of the "Run Full Node-to-Master Reconciliation" button?

It executes an authoritative audit across all employee nodes. It recovers any entries logged offline or restored from backups, validates schema headers and renewal dates, and recomputes all quota tallies and star ratings to ensure 100% data integrity without data loss.

#### Q: In emergency, how to restore database from backup created in Google Drive?

In the Admin Office Portal under "Google Drive & Node Sync Health", click **"One-Click Restore Database 🔄"**, solve the dynamic CAPTCHA security code, and enter the Master Admin Password. The system restores both Master DB and all 7 Regional Node DBs and triggers automatic reconciliation. Server CLI: `python restore_backup.py <file.zip>`.

#### National Digital Library of India (NDLI) Club Management System

Developed & Architected by **Dr. Anirban Mukherjee** • Indian Institute of Technology Kharagpur  
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